BT-202 (GS)

B.Tech., I & II Semester Examination, June 2025

Grading System (GS)

MATHEMATICS-II

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Solve: $(e^y + 1)\cos x\,dx + e^y \sin x\,dy = 0$. **7**

(b) Solve: $(D^2 - 5D + 6)\,y = 4e^x + 5$. **7**

2. (a) Solve: $x^2 \frac{d^2y}{dx^2} + 5x\frac{dy}{dx} + 4y = x \log x$. **7**

(b) Solve in series Legendre's differential equation $(1 - x^2)\,y'' - 2xy' + 2y = 0$. **7**

3. Solve $(D^2 + a^2)\,y = \tan ax$ by using the method of variation of parameters. **14**

4. (a) Eliminate the arbitrary function f from the relation $z = y^2 + 2\,f(1/x + \log y)$. **7**

(b) Solve the partial differential equation $\partial^2 z/\partial x^2 - \partial^2 z/\partial y^2 = x^2 y$. **7**

5. (a) Solve: $(y + z)\,p + (x + z)\,q = x + y$. **7**

(b) Find all values of K such that $f(z) = e^x(\cos ky + i \sin ky)$ is analytic. **7**

6. (a) Evaluate $\int$ from (0,0) to (1,1) of $(3x^2 + 4xy + i\,x^2)\,dz$ along $y = x^2$. **7**

(b) If $f(z) = 1 / [(z - 1)(z - 2)^4]$, find residue of all poles. **7**

7. (a) If $\bar{r}$ is the position vector of any point in space, then prove that the field derived from $\bar{r}$ is irrotational. (Source text partially unclear.) **7**

(b) Find the work done by the force $F = z\,\hat{i} + x\,\hat{j} + y\,\hat{k}$, when it moves a particle along the arc of the curve $\bar{r} = \cos t\,\hat{i} + \sin t\,\hat{j} - t\,\hat{k}$ from $t = 0$ to $t = 2\pi$. **7**

8. (source me ye question available nahi tha — original paper dekhein) **—**

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